

Voter models with heterozygosity selection

Probability Seminar

Organizer: Christian Gromoll & Tai Melcher

Monday, 2:00–3:00pm, Kerchof 326

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Abstract

In this talk we consider variations of the usual voter model which favor types that are locally less common. This may be understood as a selective advantage for rare types leading to more variation in the particle population. The voter models with selection considered here are dual to certain systems of branching annihilating random walks that are parity preserving. Coexistence of types in one model is related to survival of particles in the other. We consider conditions for the existence of homogeneous invariant laws in which types coexist as well as convergence to these laws. This is joint work with Jan Swart (UTIA Prague).

Original document

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